A WHITEPAPER

Nirog

AI-Assisted Prescription Intelligence and Medication Adherence for Bangladesh

A Deep-Learning Approach to Handwritten Prescription OCR, Semantic Medicine Matching, and Offline-First Medication Management

Mubtasim Fuad[1] **Md Kawsar Ahmed Kazol**[2] **Shahriar Hasan**[3]
Md. Yashrif Arifin Swoccho[4] **Md. Nazrul Islam**[5]

1 Hangzhou Dianzi University, Zhejiang, China
`mubtasimfuad2@gmail.com`

2 Khulna University of Engineering and Technology, Khulna, Bangladesh
`kazol1196295@gmail.com`

3 Hajee Mohammad Danesh Science and Technology University, Dinajpur, Bangladesh
`hasanshahriar32@gmail.com`

4 Jahangirnagar University, Dhaka, Bangladesh
`yaswoccho@gmail.com`

5 Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh
`mnnazrulislam196130@gmail.com`

A Competition Whitepaper for BCOLBD July 2026 | Version 1.0

Abstract

Adherence to prescribed medication remains one of the most persistent and costly challenges in modern healthcare, and in Bangladesh the problem is compounded by a prescription culture built almost entirely on handwritten, mixed-script documents. Doctors routinely write prescriptions that blend Bengali script with English medical terminology, Latin pharmacological abbreviations, and idiosyncratic cursive handwriting, while medicines themselves circulate under brand names issued by hundreds of local manufacturers rather than their internationally recognized generic equivalents. The result is a fragmented, error-prone chain between a doctor's ink and a patient's daily routine: medicines are misread, dosages are misremembered, refills are forgotten, and chronic conditions quietly deteriorate.

This whitepaper proposes **Nirog** (Bengali: "healthy, free from illness"), a medication-management platform designed specifically for the Bangladeshi context, whose core differentiator is an **AI-driven prescription intelligence pipeline**. The pipeline reads handwritten prescriptions using a two-pass vision-language model (DeepSeek-VL2) after OpenCV-based preprocessing, organizes the raw transcription into structured medication entries using a Bangladeshi medical abbreviation schema, and then resolves each extracted medicine name against a curated semantic knowledge base of more than 21,000 Bangladeshi branded medicines, approximately 1,700 generics, and 245 manufacturers. Resolution is performed by a four-strategy matching ensemble consisting of exact, fuzzy-phonetic, semantic vector, and LLM reranking, whose output is passed through conservative confidence tiers that either attach a medicine automatically, suggest ranked candidates for user confirmation, or explicitly ask the user when certainty is insufficient. Every user confirmation or correction is fed into a nightly active-learning loop that promotes verified spellings, aliases, and brand-to-generic mappings into the shared knowledge base, meaning the system improves with every prescription scanned.

The recognized medicines are wired into a complete adherence loop of scheduled reminders that fire on-device even when the application is offline, dose logging, refill alerts, and adherence statistics, supported by a modular FastAPI backend, a Flutter client, and a separately deployable ML service. We describe the full methodology, the data model, the AI/ML pipeline in detail, and the evaluation targets the system is designed to meet (character error rate below 8%, field extraction F1 above 0.85, medicine match top-1 accuracy above 0.90, and top-3 recall above 0.98). The framework demonstrates that a carefully engineered combination of vision-language models, multilingual embeddings, and a community-grown knowledge base can bring prescription intelligence, a capability largely reserved for structured e-prescription ecosystems, to the handwritten reality of a lower-middle-income healthcare market, with potential applications extending to pharmacy automation, insurance adjudication, and public-health surveillance.

Keywords: Medication Adherence, Optical Character Recognition, Vision-Language Models, Semantic Matching, Multilingual Embeddings, Health Informatics, Bangladesh, Bangla NLP

Contents

Chapter 1

Introduction

The digital transformation of healthcare has made remarkable progress in recent decades, yet a fundamental disconnect persists between how prescriptions are written and how they are consumed. In high-income markets, electronic prescribing (e-prescribing) has largely replaced handwritten notes, structuring the doctor's intent into machine-readable orders that pharmacy and patient systems can consume directly. In much of the developing world, however, the handwritten prescription remains the default instrument of the physician's craft, and nowhere is this more consequential than in Bangladesh, a country of over 170 million people with a rapidly growing chronic disease burden and one of the most active pharmaceutical industries in South Asia [1].

Bangladeshi prescriptions are written in a distinctive and demanding register. A single prescription may interleave Bengali script, English drug names and dosages, Latin-derived frequency abbreviations such as *OD* (once daily), *BD* (twice daily), *HS* (at night), *PC* (after food), and *AC* (before food), numerals written in Bengali glyphs, and the hurried cursive of a physician seeing dozens of patients an hour. The medicines themselves compound the difficulty: Bangladesh's pharmaceutical industry produces more than 21,000 branded formulations of roughly 1,700 generic molecules across 245 manufacturers [1], so a single molecule such as paracetamol may appear under dozens of brand names such as *Napa*, *Ace*, *Paracip*, and many more, none of which share obvious orthographic similarity.

The downstream consequence is that a patient leaving a consultation with a handwritten slip of paper must either decipher it themselves, rely on the attending pharmacist's interpretation, or abandon parts of the regimen entirely. Studies consistently show that roughly half of patients in developed markets take their chronic medications correctly, with adherence substantially worse in low- and middle-income countries where the prescription itself becomes a barrier to correct use [2, 3]. Mobile applications have been shown to measurably improve adherence [4, 5], but the commercial solutions available today assume a clean, manual data entry workflow and provide no capability to read the prescription at all, leaving the most error-prone step, transcription, entirely in the patient's hands.

This whitepaper presents Nirog, a medication-management platform that closes that gap end to end. Its central claim is that the handwritten Bangladeshi prescription, long considered hostile territory for automation, can be read, structured, and reconciled against the local drug landscape using a pipeline that combines classical computer-vision preprocessing, a large vision-language model, and a semantic knowledge base grown through active learning from its own users.

1.1 Problem Statement

The problem decomposes into four mutually reinforcing challenges, each of which existing tools address at most partially.

Ambiguity of the handwritten source. Doctors' handwriting is inherently variable, and medical handwriting is among the most challenging cases because it is produced under time pressure, mixes scripts and codes, and contains abbreviations that are meaningful only within a narrow professional culture. Conventional off-the-shelf OCR engines, trained predominantly on clean printed text in Latin scripts, produce character error rates that make automatic medicine extraction unreliable, and errors in this domain are safety-critical: mistaking "BD" for "OD", or "5 mg" for "50 mg", changes a therapeutic dose.

The brand-to-generic gap. Patients encounter medicines under brand names; clinical meaning lives at the generic level. No widely available tool bridges the Bangladeshi brand landscape of the thousands of local brands and their generic equivalents, so even a perfect transcription would leave a patient without context: which class of drug is this, what is its purpose, is it interchangeable with the brand the pharmacy dispensed?

Fragmented adherence practice. Medicines prescribed in a consultation must then be remembered, scheduled, taken, refilled, and reviewed over weeks and months. Existing reminder applications address scheduling but not the acquisition of medicines from the prescription, forcing users to manually re-enter every drug, dosage, and frequency, precisely the step where transcription errors enter the record.

Absence of a continuously improving data substrate. Handwriting quality, brand-name diversity, and abbreviation usage differ by region, specialty, and individual physician. A static system cannot adapt; a system with no learning loop will perpetually misrecognize the same names and miss the same aliases, eroding user trust over time.

Together, these gaps describe a coherent design target: a system that reads the prescription as written, structures it with respect for local medical conventions, reconciles it with the national brand landscape, and grows smarter from every confirmed or corrected reading.

1.2 Project Objectives

Nirog pursues five primary objectives that map the problem statement to concrete system capabilities.

Objective 1, Prescription digitization. Build a robust OCR pipeline capable of transcribing handwritten Bangladeshi prescriptions with a character error rate below 8%, handling the mixed Bangla/English script, Latin abbreviations, and cursive handwriting that characterize the domain.

Objective 2, Structured medication extraction. Convert raw transcription into structured medication entries with name, dosage amount and unit, frequency with daily slots, duration, route, and instructions, with field extraction F1 above 0.85, respecting the abbreviation conventions of Bangladeshi medical practice.

Objective 3, Semantic medicine resolution. Reconcile each extracted medicine name with the Bangladeshi drug landscape by matching it against a curated knowledge base of branded medicines, generics, and manufacturer aliases, achieving top-1 match accuracy above 0.90 and top-3 recall above 0.98.

Objective 4, Adherence automation. Wire recognized medicines into a complete adherence loop of on-device scheduled reminders that fire without network connectivity, dose logging with taken/missed/skipped statuses, threshold-based refill alerts, and adherence statistics, so that scanning a prescription is the only manual work the patient performs.

Objective 5, Active-learning knowledge growth. Convert every user confirmation, correction, and new-medicine creation into permanent growth of the medicine knowledge base through a nightly promotion loop and an administrative verification workflow, so the system improves continuously with real usage.

1.3 Significance

The proposed platform addresses the challenge at a level of integration that neither component technology achieves alone, with significance across four dimensions.

Patient outcomes. Adherence interventions are among the highest-value improvements in chronic care: a meta-analysis of mobile applications in adults found statistically significant adherence gains [4], and every study in a 2020 BMJ Open systematic review reported efficacy [5]. By removing the transcription barrier, Nirog makes such interventions available to patients whose regimens currently live only on paper.

Accessibility and equity. The platform is designed to be initially fully free, targeting near-zero marginal cost per scan through open-weight models and self-hosted components, which matters in a market where out-of-pocket expenditure dominates healthcare financing. A single pipeline serves patients, caregivers managing family members' medications, and elderly users through multi-profile support.

Localization as a moat. The system's hardest assets, namely the Bangladeshi abbreviation schema, the brand-generic mapping, the 21,000-medicine corpus, and the mined alias dictionary, are specific to Bangladesh and difficult for generic international tools to replicate, while simultaneously being the features most valuable to the intended users [1].

Data infrastructure for healthtech. The verification workflow, the annotated prescription evaluation set, and the accumulated feedback corpus constitute reusable health-data infrastructure: the same resolved brand-generic graph can serve pharmacy point-of-sale, telemedicine documentation, insurance adjudication, and pharmacovigilance downstream.

Chapter 2

Literature and Domain Review

The design of Nirog draws on four bodies of prior work: medication adherence interventions, handwritten medical prescription recognition, multilingual and Bangla text recognition, and semantic entity resolution. This section surveys each and identifies the gaps the framework addresses.

2.1 Medication Adherence and Mobile Health Interventions

A large body of clinical literature establishes that mobile applications can improve medication adherence. A content analysis of adherence apps in JMIR mHealth identified over fifty applications with heterogeneous feature sets, most combining reminders with medication lists [6]. A meta-analysis published in the Journal of Managed Care & Specialty Pharmacy found that app-based interventions significantly improved adherence in adults [4], a 2019 systematic review in *JMIR mHealth and uHealth* reported that users found medication apps easy to use and useful [2], and a 2020 BMJ Open systematic review concluded that all nine app interventions it examined demonstrated efficacy, with five achieving statistical significance [5]. More recent evidence remains consistent: a 2025 evaluation of mobile apps reported that all fourteen studied interventions improved adherence and ten randomized controlled trials showed statistically significant improvement [7].

The persistent limitation of this literature, however, is that every examined application treats **medication data entry as a given**; the user, a caregiver, or a pharmacist must manually assemble the regimen. No study addresses the transcription of handwritten prescriptions into digital regimens, which our experience suggests is precisely the step where users disengage and where error enters. The gap is not a reminder engine; it is the bridge between paper and the engine.

2.2 Handwritten Medical Prescription Recognition

Prescription recognition has attracted sustained research precisely because of its difficulty and impact. A hybrid deep-learning approach combining multilayer perceptrons with convolutional feature extractors was proposed for doctors' handwritten prescriptions in the *Journal of Data Governance* [8], and a deep-learning framework pairing OCR with natural language processing was reported for real-time prescription reading [9]. Earlier applied work includes DocPresRec, which applies deep learning to doctors' handwritten prescriptions within clinical documentation research [10], and multi-engine OCR pipelines augmented with deep contextual NLP that propose intelligent digitization of medical prescriptions [11]. Open-source efforts such as the Donut-based Medical-Prescription-OCR project demonstrate end-to-end structured extraction from prescription images [12].

Three patterns emerge from this literature. First, end-to-end deep models (transformer-based recognizers) now dominate over classical CRNN pipelines for this domain. Second, the most reliable systems treat recognition and *interpretation* as separate stages, so that recognition should be faithful to the ink, while interpretation applies domain knowledge (abbreviation expansion, dosage normalization). Third, published work concentrates on English-language prescriptions from high-income settings; prescriptions mixing Bengali script with English medical shorthand remain substantially under-addressed.

2.3 Multilingual and Bangla Text Recognition

Bangla handwriting recognition presents distinct structural challenges: the script's compound characters (juktakkhor) form complex connected glyphs that defeat character-segmentation approaches, and mixed-script documents interleave Bangla with Latin words mid-line. Recent work includes a hybrid Bangla handwritten OCR pipeline combining YOLO-based detection with recognition components [13], a bilingual Bangla OCR system aimed at rural empowerment [14], a dedicated Bangla Handwritten Character Recognition dataset with 583 character classes [15], and multi-head attention architectures for Bangla handwritten text recognition [16]. Classical research on mixed Bangla-English recognition dates back at least to 2017 [17].

These results confirm that the subproblems are individually tractable but have not been assembled into a prescription-aware pipeline. Nirog's organizer step, covering section detection, two-column merging, Bangla numeral normalization, and abbreviation expansion, is, to our knowledge, a novel composition of these components aimed specifically at the prescription document type.

2.4 Semantic Entity Resolution for Medicines

Medicine entity resolution, that is, deciding that "Napa 500", "napa tab", the local brand alias for Napa, and "paracetamol 500 mg" all denote the same generic molecule, is a named-entity and record-linkage problem with domain-specific texture. The Bangladeshi market amplifies every standard difficulty: hundreds of manufacturers, dozens of brands per generic, frequent user misspellings, and transliteration between scripts. Semantic search approaches using transformer embeddings (BERT and successors) have demonstrated strong performance on paraphrase and synonym detection in clinical text, and multilingual embedding models such as BGE-M3, which support over 100 languages including Bangla with combined dense, sparse, and multi-vector retrieval [18, 19], make cross-script semantic matching practical without per-language model training.

2.5 Research Gaps and Opportunities

The preceding survey reveals four specific gaps. **(G1)** No integrated system connects handwritten prescription OCR to an adherence loop; recognition research stops at structured text, and adherence research starts at structured text. **(G2)** Bangladeshi mixed-script prescriptions are under-served: existing prescription-OCR literature addresses English-language documents, and Bangla OCR literature addresses generic documents rather than the prescription register. **(G3)** Local drug-landscape knowledge, namely the brand/generic/manufacturer graph for a specific national market, is absent from general-purpose tools and constitutes the decisive practical asset. **(G4)** Knowledge-base growth from user feedback, that is, the mechanism by which a system compensates for the long tail of physician handwriting styles and local brand variants, is rarely built as a first-class component. Nirog's design addresses all four gaps simultaneously, which constitutes its principal contribution.

Chapter 3

Methodology

Our framework employs a methodology that integrates on-device mobile engineering, a modular backend, and a deep-learning OCR and matching pipeline to create a complete prescription-to-adherence system. This section details the architectural design, the technical implementation, and the operational workflow that enables the system to function effectively at scale.

3.1 System Architecture

The proposed framework consists of five primary components that work in harmony to provide a complete medication-management solution, arranged in a **client–gateway–modular-monolith** topology with a separately deployable machine-learning service.

1. **Flutter Mobile Application.** The primary client, available for Android and iOS, implements offline-first medication management: local scheduled notifications for dose reminders, a local database cache of medicines and dose logs, and manual medicine entry. The app remains fully functional without network connectivity, reconciling with the server when a connection is available.

2. **API Gateway and Load Balancer.** A TLS-terminating edge layer that routes REST and WebSocket traffic, enforces rate limits, and dispatches scan requests between synchronous and asynchronous handling modes.

3. **Modular Monolith Backend (FastAPI).** A Python backend organized into strict domain modules comprising `users`, `medicines`, `reminders`, `ocr`, `notifications`, and `admin`, that communicate exclusively through service interfaces. This pattern keeps early-stage operation simple while keeping every boundary clean enough to extract into microservices without re-engineering [20].

4. **ML Service.** A separately deployable FastAPI service hosting the complete OCR pipeline: image preprocessing, two-pass vision-language-model recognition, structured extraction, and semantic medicine matching. It consumes scan jobs from a Redis queue executed by a Celery worker pool and is designed to scale vertically or onto GPU hardware, independently of the transactional web tier.

5. **Knowledge Base and Data Layer.** PostgreSQL provides the transactional schema and full-text/trigram medicine search; a vector index (pgvector initially, Qdrant at scale) hosts multilingual embeddings of the medicine corpus; Redis provides caching and job queuing; and object storage (MinIO/S3) retains prescription images with SHA-256 content addressing that doubles as cache key and audit identifier.

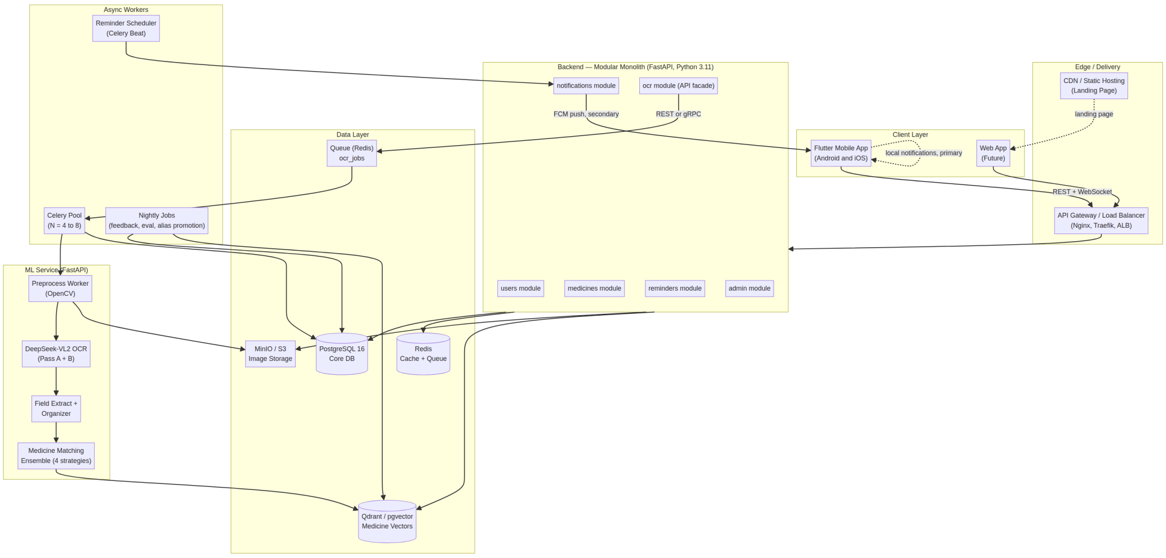

Figure 3.1: High-level system design showing the five components and their interactions.

3.2 Technical Implementation

3.2.1 AI-Powered Prescription OCR

The OCR pipeline is the technical heart of the framework and is implemented as a staged chain of independently testable components. Image preprocessing (deskewing, denoising, CLAHE contrast enhancement, adaptive thresholding, shadow removal, and resizing into the model's optimal input range) is performed with OpenCV in a worker process. Recognition is delegated to **DeepSeek-VL2**, a mixture-of-experts vision-language model [21], executed in two passes: a faithful transcription pass that copies the ink verbatim with per-line confidence scores, followed by a structured extraction pass that normalizes the transcription into a medication schema and expands Bangladeshi medical abbreviations. Semantic medicine matching then resolves each extracted name against a corpus of more than 21,000 Bangladeshi branded medicines embedded with **BGE-M3**, a multilingual model that natively supports Bangla [18, 19]. A detailed treatment follows in Section 4.3.

3.2.2 Mobile Client and Backend Integration

The client follows Flutter's official layered MVVM architecture [22], with a UI layer, a data layer of repository implementations over a remote API client (Dio) and local persistence (Drift/SQLite), and an optional domain layer of use cases. The backend registers every medicine with its normalized frequency, from which the client derives zone-aware local notification schedules that fire on the device regardless of network state; the server maintains the authoritative schedule and receives snooze and dose-log events so that adherence statistics and caregiver views stay consistent across devices.

3.3 System Flow

The complete system workflow follows a carefully designed sequence of steps that ensures both accuracy and efficiency. The core of our framework is a prescription intelligence pipeline that

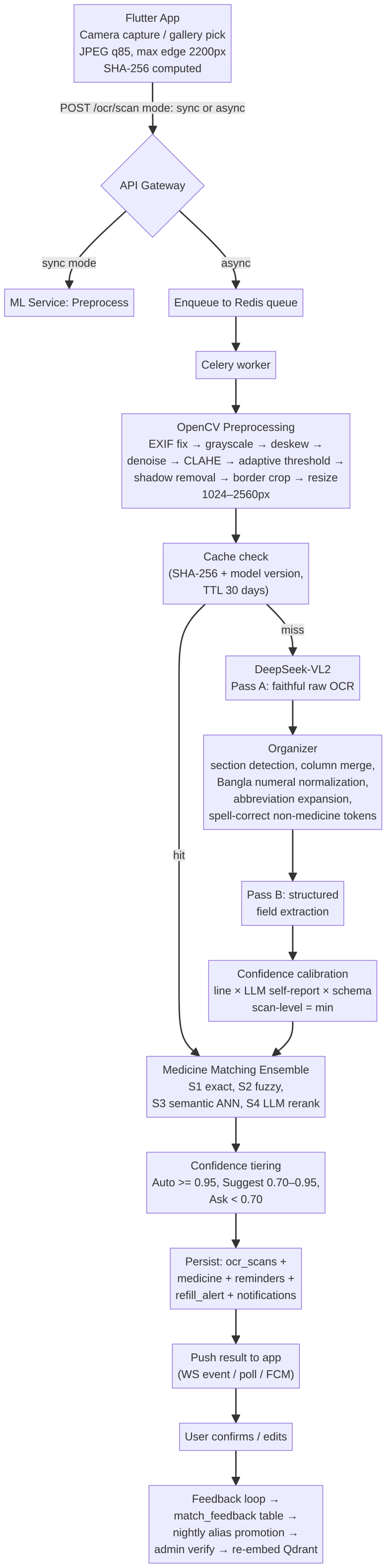

Figure 3.2: The full OCR and medicine-matching pipeline from image ingestion to persistence.

seamlessly connects the patient's scan action, backend AI processing, semantic resolution, and the creation of a persistent adherence regimen.

3.3.1 Frontend to Backend Pipeline

The pipeline begins at the front end, where the user captures a prescription photograph with the camera or selects one from the gallery. The client compresses the image to JPEG at quality 85 with a maximum edge of 2200 pixels, computes its SHA-256 digest for caching and deduplication, and posts the image to the gateway with a handling mode. **Synchronous** mode blocks for at most twelve seconds and returns the fully structured result directly, suited to short, single-tablet scribbles. **Asynchronous** mode returns a `scanId` immediately, enqueues the job, and delivers the completed result through a WebSocket event stream, with a push notification fallback when the application is backgrounded. Throughout processing, the application surfaces live progress and, on completion, the recognized medicines with their confidence tiers and a confirmation interface.

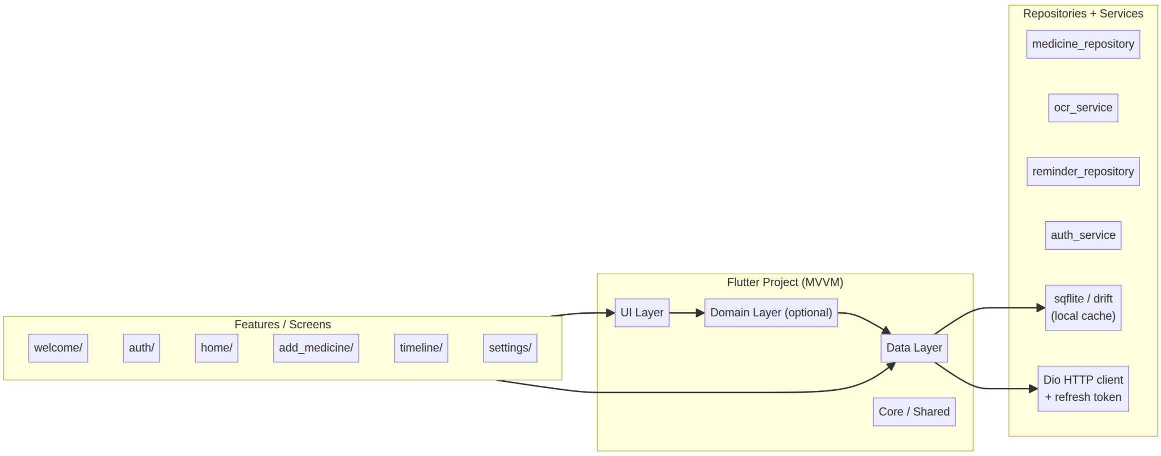

Figure 3.3: The client architecture through which scan capture, local notification scheduling, and dose logging are orchestrated.

3.3.2 Backend AI Processing Pipeline

Once content reaches the backend, it enters the core processing pipeline. An OpenCV preprocessing worker normalizes the image (orientation fix, grayscale, deskew, denoising, contrast enhancement, thresholding, shadow removal, border cropping, and resize to the 1024–2560 pixel band). The vision-language model then performs Pass A (verbatim transcription with per-line confidence) and Pass B (structured extraction with abbreviation expansion). The organizer hybrid step sections the document into clinical regions, merges two-column layouts, normalizes Bangla numerals, and expands abbreviations from a hot-reloadable Bangladeshi medical abbreviation table. Every pipeline task is keyed by scan identifier plus stage, so retries never double-persist; failed tasks fall to a dead-letter queue for inspection.

3.3.3 Semantic Resolution and Persistence Process

For content that passes extraction, the system initiates semantic resolution. Each extracted medicine name is scored by a four-strategy ensemble, beginning with an exact case-insensitive and accent-

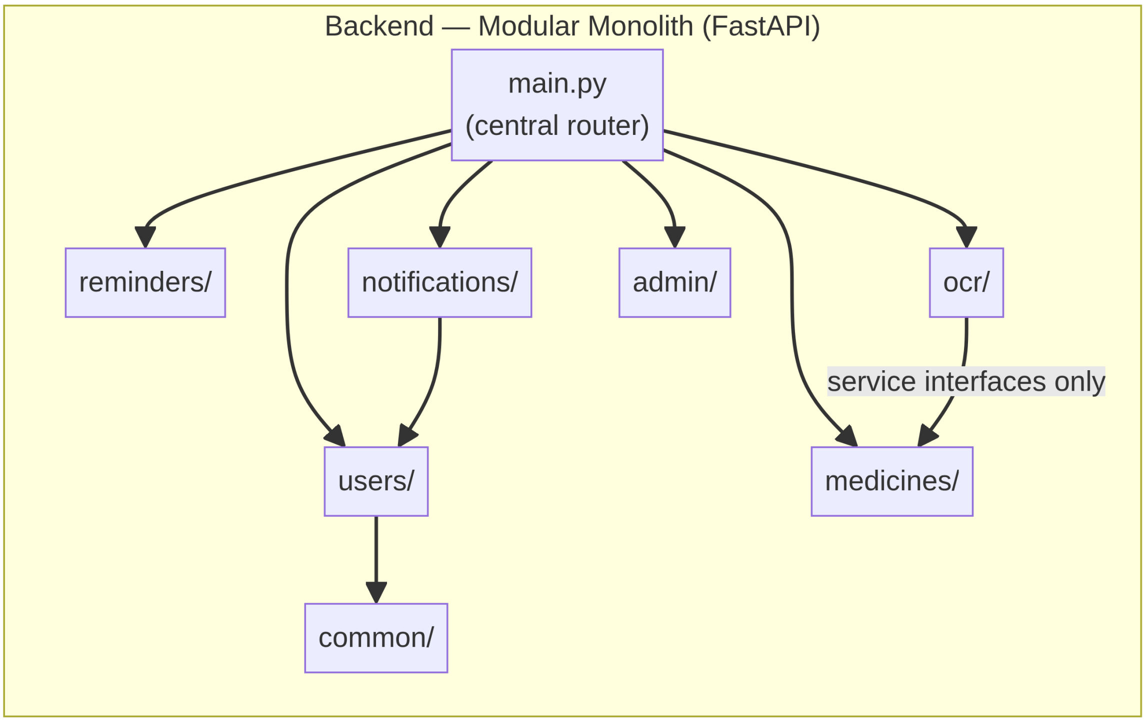

Figure 3.4: The backend modular monolith, with domain modules communicating through service interfaces only.

folded match against names, generics, and aliases; fuzzy string distance with phonetic matching via Metaphone; approximate-nearest-neighbor cosine similarity over BGE-M3 embeddings in the vector index; and LLM reranking of the merged candidate set by the vision-language model. The merged scores pass through confidence tiers: medicines at or above 0.95 are attached automatically; those between 0.70 and 0.95 are presented with the top three ranked candidates and the first pre-selected; those below 0.70 present five candidates plus an explicit “None of these, add a new medicine” path, so the system never silently fails. Accepted medicines are written transactionally together with reminder schedules derived from the normalized frequency, and any new unverified medicine enters the administrative verification queue, after which it is re-embedded and served to all future users.

3.3.4 Complete Workflow Integration

The integrated pipeline operates as a seamless system:

1. **Capture and submission.** The user photographs the prescription; the client compresses, hashes, and submits it, choosing synchronous or asynchronous handling.

2. **Preprocessing and transcription.** The preprocessing worker normalizes the image; the model transcribes it faithfully in Pass A.

3. **Structuring.** The organizer sections and cleans the text; Pass B emits structured medication entries with field-level confidence.

4. **Resolution.** The four-strategy ensemble scores each medicine name; the confidence tier determines the user-facing action; generic alignment surfaces the molecule identity.

5. **Confirmation.** The user accepts, edits, or rejects each suggestion; rejections are recorded as feedback.

6. **Regimen creation.** Accepted medicines are persisted with reminders, refill alerts, and dose-logging scaffolding; local notifications are scheduled immediately.

7. **Learning.** Confirmations and corrections accumulate in the feedback substrate; a nightly job promotes consensus mappings into aliases and flags repeated new-medicine creations for administrative verification.

3.4 User Interface and Experience

The front end provides an intuitive experience organized around the patient's daily routine rather than around data entry. The onboarding sequence explains the OCR workflow in three slides before the first scan. The home screen presents today's medicines as a checklist with adherence statistics, streamed from the local database. Medicines are added through a dual path of a manual form for ad-hoc entries and the OCR scanner for prescription capture, with the same search completion powered by the shared knowledge base in both cases. A timeline and calendar view supports retroactive dose logging, and settings expose preferences for language (English/Bangla), theme, timezone, snooze duration, and weekly digest delivery.

The confirmation interface is where the confidence tiering becomes visible to the user. A green "matched" badge indicates an automatically attached medicine; a yellow suggestion card shows ranked alternatives the user may reselect; and a red "verify" card presents candidates plus the explicit escape path to add a new medicine. This graduated transparency is deliberate: it keeps the common case frictionless while never hiding uncertainty from the user, which in a health context is the fastest route to lost trust. Administrative workflows for verifying pending medicines, reviewing the alias queue, and monitoring pipeline metrics live in a separate admin panel so that clinical-quality data curation can scale alongside user growth.

Chapter 4

Implementation Details

This chapter specifies the concrete implementation of the framework's core components: the application and backend architecture, the data model and medicine knowledge base, the AI/ML pipeline, and the reliability, security, and privacy safeguards.

4.1 Application and Backend Architecture

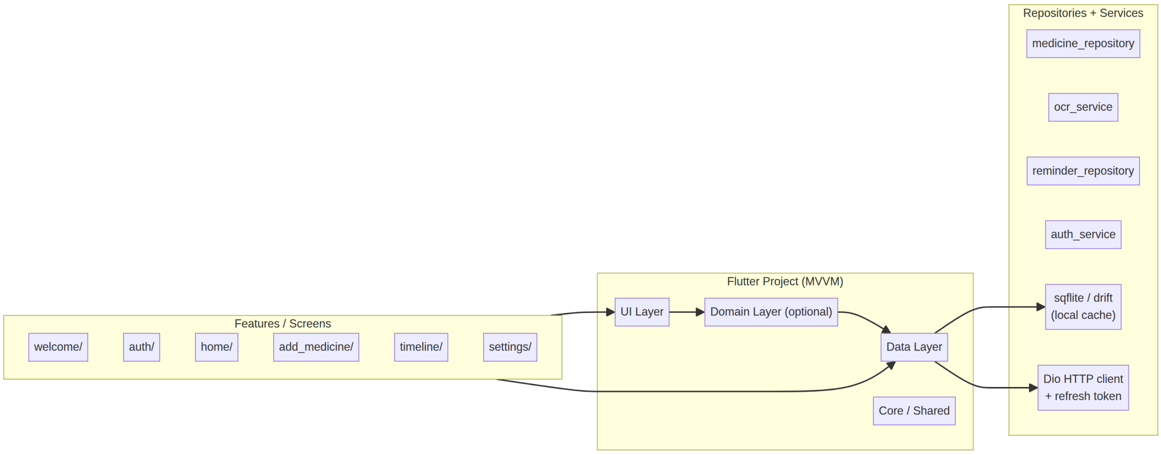

Figure 4.1: The Flutter application's layered structure, from views and view models to repositories and remote/local data sources.

The mobile application is a Flutter codebase following the official layered architecture guide [22], organized by feature (`auth`, `home`, `medicines`, `reminders`, `ocr_scan`, `settings`, `notifications`), each owning a presentation layer of views, view models, and widgets that depend on repository interfaces defined in the domain layer. Authentication combines email registration with OAuth through JWT access tokens and refresh tokens persisted in secure storage. The most consequential client-side decision is the **dual-track notification strategy**: because push notification delivery cannot be guaranteed, and Firebase Cloud Messaging's own documentation makes no delivery commitment [23]; dose reminders are scheduled as zone-aware **local notifications** computed from the medicine's normalized frequency at accept time, firing on-device even when the application is closed and the device is offline; server push (FCM) is reserved for non-critical nudges such as scan completion, refill alerts, and weekly digests, where occasional loss is acceptable [24].

The backend is a FastAPI modular monolith (Python 3.11) whose six domain modules each define models and services and communicate exclusively through service interfaces. For example, the notifications module requests medicine details through `MedicineService.get_medicine()` rather than importing the `Medicine` model directly, keeping each module framework-free and portable [20]. Cross-cutting mechanisms include an `isVerified` visibility model (verified medicines are searchable

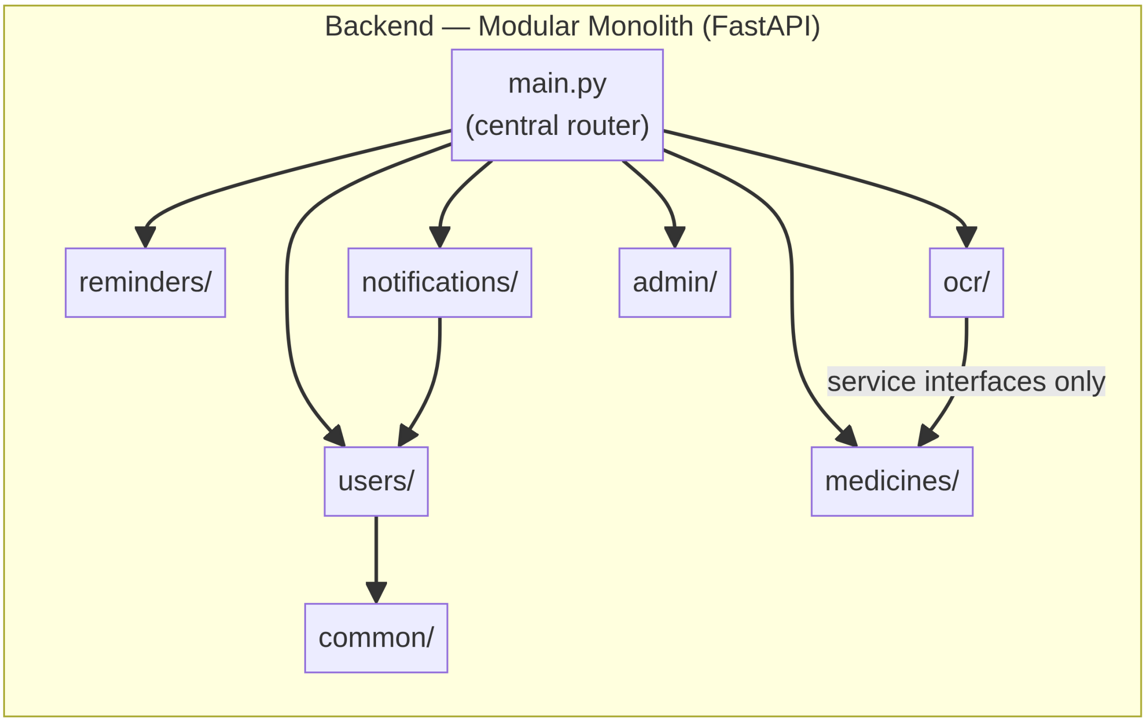

Figure 4.2: The backend modular monolith, with domain modules communicating through service interfaces only.

by all users; unverified, user-added medicines are visible only to their creator until administrative review), a hybrid medicine search combining exact B-tree, trigram (`pg_trgm`), and vector (HNSW) retrieval so that client search and OCR matching share one source of truth, and idempotent asynchronous jobs keyed by `scanId + stage` with exponential-backoff retries and a dead-letter queue.

4.2 Data Model and Medicine Knowledge Base

The domain model centers on `User` → `UserProfile` → `Medicine`, with `Reminder`, `DoseLog`, `RefillAlert`, `RefillHistory`, `AdherenceStats`, and `OCRScan` as dependent aggregates, and a shared `Notification` event entity triggered by reminders, refill alerts, and dose logs alike. The ML-specific tables extend this core: `ocr_scans` retains the full audit record of every scan including raw text, extracted fields, matches, confidence, and the prompt and model versions that produced them; `medicine_aliases` accumulates aliases from user corrections, administrative curation, and the base dataset; `match_feedback` records every `(input_token, chosen_medicine_id)` decision as the substrate of active learning; and `pending_medicines` queues unverified user-added medicines for review.

The semantic knowledge base merges four sources into a single corpus: the **Assorted Medicine Dataset of Bangladesh**, containing more than 21,000 branded medicines, approximately 1,700 generics, 245 manufacturers, and 123 dosage forms, CC0-licensed and sourced from medex.com.bd [1], verified user-added medicines, a generic-to-brand mapping table, and curated aliases and mined misspellings harvested from accepted user corrections. Each medicine is embedded at 1024 dimensions with **BGE-M3** [18, 19], self-hosted with a managed embedding fallback, as the concatenation of

Table 4.1: Core domain entities and their roles.

Entity	Core Fields	Notes
`User`	id, email, passwordHash, name, dob, role, createdAt	Registration, OAuth, account deletion cascade
`UserPreferences`	theme, language, timezone, snoozeDuration, digests	1:1 with user
`UserProfile`	name, relationship, isDefault	Multi-member household management
`Medicine`	name, dosage, unit, frequency, dates, stockCount, refillThreshold	Owns reminders, dose logs, refill alerts, stats
`OCRScan`	imageUrl, extracted fields, status, timestamps	0..1 creates a Medicine
`Reminder`	scheduleType, reminderTimes[], snoozeMinutes, nextTrigger	Triggers notifications
`DoseLog`	scheduledTime, actualTime, status, isRetroactive	Adherence measurement
`RefillAlert`	threshold, alertSentAt, isAcknowledged	1:1 per medicine
	ML-Specific Tables	
`ocr_scans` (ML)	raw_text, extracted, matches, confidence, prompt/model versions	Full audit trail
`match_feedback` (ML)	scan_id, input_token, chosen_medicine_id, is_new	Active-learning substrate
`pending_medicines` (ML)	name, suggested_generic, occurrences, status	Admin verification queue

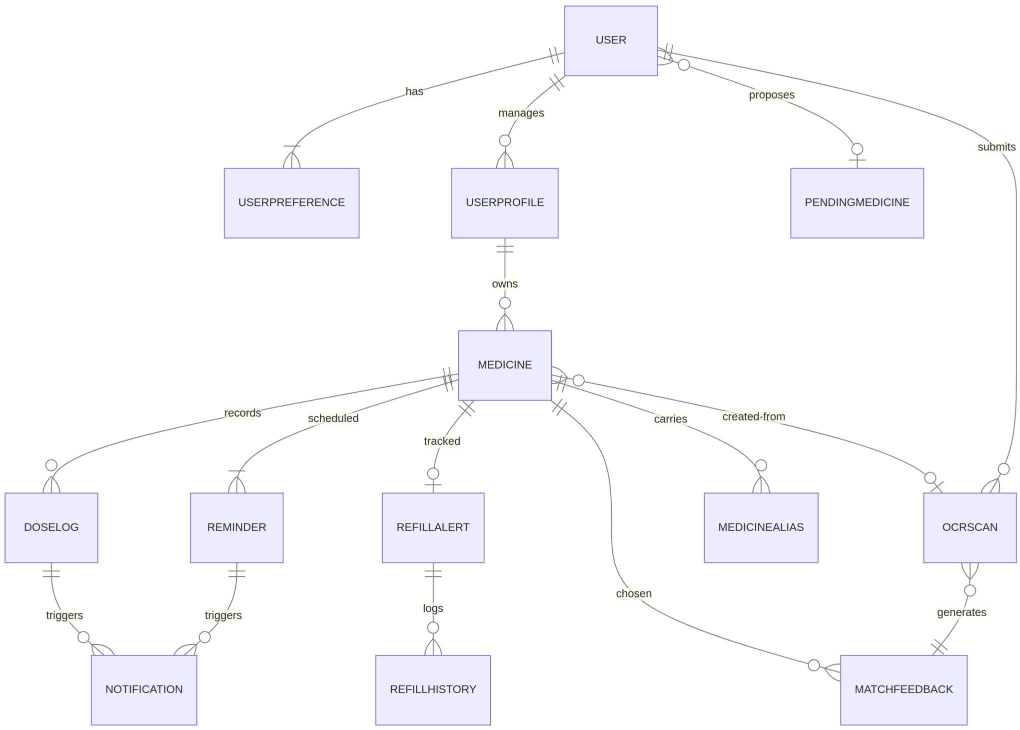

Figure 4.3: Entity-relationship diagram showing core domain entities and their relationships.

name, generic, strength, form, and alias list, using HNSW approximate-nearest-neighbor retrieval complemented by exact B-tree and trigram fallback indexes. The `User` $\rightarrow$ `Medicine` visibility rules, the shared search index, and the feedback-driven alias promotion together ensure that the knowledge base is simultaneously authoritative, searchable, and self-improving.

4.3 AI/ML Pipeline

The pipeline executes in seven stages: ingestion, preprocessing, two-pass OCR, text organization, semantic matching, confidence tiering with generic alignment, and persistence with feedback, each implemented as a discrete, testable component chained through Celery tasks in the asynchronous path.

Ingestion. The client compresses captures to JPEG quality 85 (maximum edge 2200 px), hashes them, and posts to `/ocr/scan`. The gateway enforces a twelve-second blocking budget for synchronous scans and immediately returns a `scanId` for asynchronous jobs, whose raw images are retained in object storage at `ocr/{userId}/{date}/{sha256}.jpg` for audit while only the preprocessed image reaches the model.

Preprocessing (OpenCV). A worker applies, in order: EXIF orientation correction, grayscale conversion, deskew via Hough transform on text contours, non-local means denoising, CLAHE contrast enhancement, adaptive Gaussian thresholding tuned for handwriting, LAB-channel shadow removal, border cropping to the largest contour, and resize into the 1024–2560 pixel band. Multi-page prescriptions are optionally split into region-of-interest tiles.

Two-pass recognition (DeepSeek-VL2). Pass A instructs the model to transcribe exactly as written, applying no spelling correction and no abbreviation interpretation, returning per-line text with confidence and a language hint (`en | bn | mixed`). Pass B operates on that output alone, expanding Bangladeshi medical abbreviations (`OD`, `BD/BID`, `TD/TID`, `QID`, `HS`, `PC`, `AC`, `STAT`) and normalizing mixed Bangla/English terms into the structured schema: medicine name, dosage amount and unit, frequency with times-per-day and slots, duration, route, instructions, and per-entry confidence. **TrOCR** serves as a fallback ensemble for low-confidence regions [25]. Cache keys combine the preprocessed image hash with the model version at a 30-day TTL, so repeat and similar prescriptions skip both passes entirely.

Organizer. Before Pass B, a hybrid regex-plus-LLM step buckets lines into clinical sections (`PATIENT_INFO | RX | MEDICATIONS | INSTRUCTIONS | DOCTOR | DATE`), merges two-column layouts by y-coordinate, normalizes Bangla numerals, expands abbreviations from a hot-reloadable YAML table, and applies spell correction only to non-medicine tokens, preserving raw medicine names for the matching layer.

Matching ensemble. Every extracted name passes through four strategies merged by a hot-reloadable weight configuration:

Table 4.2: Four-strategy medicine matching ensemble.

Strategy	Method	Score Semantics
S1: Exact	Case-insensitive, accent-folded match against name/generic/aliases	1.00 on hit
S2: Fuzzy	Normalized Levenshtein, Jaro-Winkler, Metaphone; maximum of three	0–1
S3: Semantic	Vector ANN top-K=10, cosine similarity	0–1
S4: LLM rerank	Model reranks the S1–S3 candidate union by likelihood	0–1

The final score is computed as:

$$\text{final\_score} = 0.40 \cdot \text{semantic} + 0.25 \cdot \text{fuzzy} + 0.25 \cdot \text{llm\_rerank} + 0.10 \cdot \text{exact\_bonus} \qquad (4.1)$$

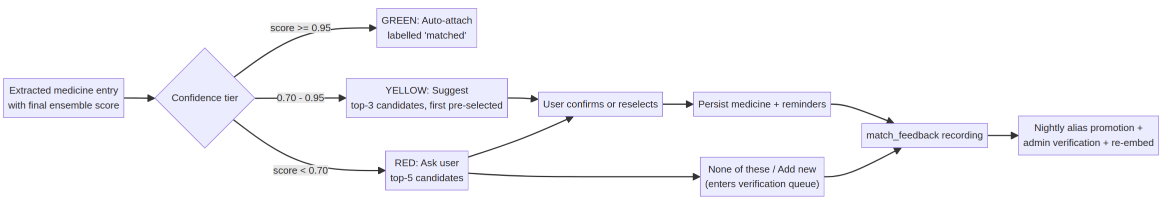

Figure 4.4: The confidence tier decision flow: every uncertainty escalates to the user, and every decision feeds the learning loop.

Confidence tiers. Field-level confidence composes the source-line confidence, the model's self-reported confidence, and schema-validation pass/fail; scan-level confidence is conservatively the minimum across entries, because an inflated confidence that silently attaches the wrong medicine erodes trust faster than any latency issue. The tiered action matrix then decides the user experience, auto-attaching at ≥ 0.95, ranked suggestion at 0.70–0.95, explicit ask below 0.70 with an escape path, and identity resolution is followed by generic lookup and, for unmatched names, creation of an unverified medicine pending review.

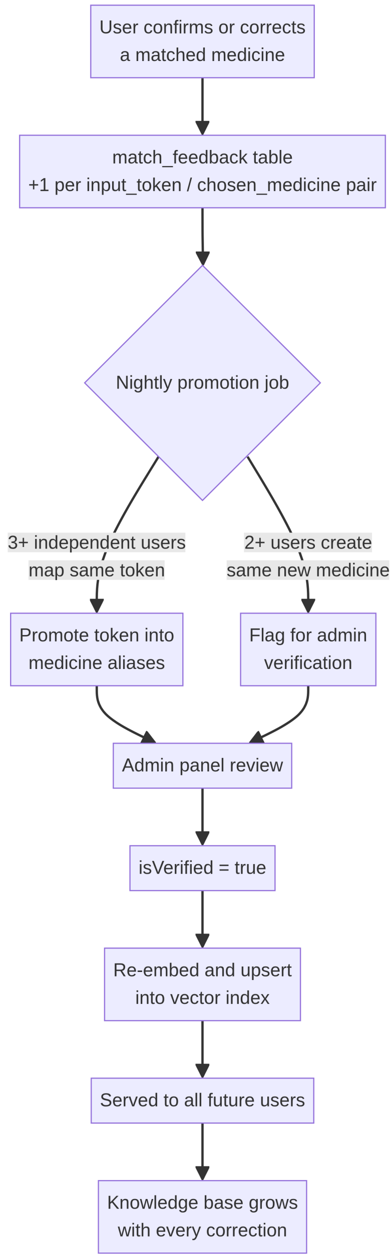

Figure 4.5: The active-learning feedback loop in which user corrections are aggregated nightly and, after administrative verification, re-embedded into the shared knowledge base.

Active learning. Each confirmation or rejection increments its (`input_token, chosen_medicine_id`) pair in `match_feedback`; a nightly job promotes tokens that three or more independent users map to the same medicine into that medicine's aliases, flags repeated near-duplicate new medicines for verification, and administrative approval triggers re-embedding so the correction is served to all future users. The loop converts every correction into permanent knowledge-base growth.

4.4 Reliability, Security, and Privacy Considerations

Prescription images and extracted medication records are personal health information and are treated accordingly even ahead of formal regulatory obligations. Authentication uses short-lived JWT access tokens with refresh tokens in secure client storage. Images and extracted text are encrypted at rest (server-side encryption on object storage, encrypted database volumes) and TLS-encrypted in transit; every endpoint enforces row-level `userId` isolation rather than trusting client-supplied identifiers, and account deletion cascade-wipes scans, medicines, and feedback records. Rate limiting applies per-user and per-IP on scan endpoints, and a per-user daily token cap plus a circuit breaker bound LLM expenditure; cache hits are the largest cost lever, since every cached or similar prescription avoids both recognition passes.

On the reliability side, the dual-track notification strategy guarantees that life-critical dose reminders never depend on network state, and the end-to-end workflow is instrumented with OpenTelemetry tracing across preprocessing, recognition, and matching, with dashboards tracking the confidence histogram, acceptance rates, dead-letter depth, and dollar spend, and a drift alert paged when the auto-match rate regresses by more than ten percent week-over-week.

Chapter 5

Implementation Status and Development Roadmap

5.1 Current Development Phase

The project has completed its Phase 2 feasibility study and full system design. The core product proposition of OCR-driven medicine extraction followed by semantic matching and adherence automation, has been validated at the design level: the knowledge base corpus is identified and licensed (CC0), the model candidates are benchmarked on paper, the data model and API surface are specified, and the evaluation methodology is defined. The remaining work is implementation across six phased releases.

5.2 Development Milestones

Table 5.1: Development phases, scope, and exit criteria.

Phase	Scope	Exit Criteria
0: Foundation	Repository scaffold, containerized local stack, authentication, landing page (live)	Authenticated API plus application shell
1: Core loop	Manual medicine entry, local reminders, dose logging, adherence statistics, backend sync	A user can manage medicines without OCR
2: OCR v1	Synchronous and asynchronous scan endpoints, preprocessing, two-pass recognition, knowledge-base seed from the Kaggle corpus	CER $<$ 8%, top-3 recall $>$ 0.95 on the evaluation set
3: Matching v1	Four-strategy ensemble, confidence tiers, generic alignment, administrative verification panel	Auto-match $>$ 50%, correction rate $<$ 20%
4: Feedback loop	`match_feedback` accumulation, nightly alias promotion, pending-medicines queue	Aliases growing weekly from real users
5: Hardening	Refill tracking, multi-profile, weekly digests, observability, cost guardrails, privacy review	All non-functional targets met; load-tested

The application data layer requires no OCR to be built, and the ML service can be developed and evaluated offline against the curated prescription set, so Phases 0–1 and Phase 2 proceed in parallel.

5.3 Expected Outcomes

By the end of the roadmap, the framework expects to deliver a production system in which a patient photographs a handwritten prescription and, within seconds, holds a structured medication regimen with scheduled reminders, refill awareness, and adherence measurement, without having typed a single drug name. The evaluation harness defines the quality bar quantitatively:

Table 5.2: Quantitative evaluation targets.

Metric	Target
OCR character error rate	$< 8\%$
Field extraction F1	> 0.85
Medicine match, top-1 accuracy	> 0.90
Medicine match, top-3 recall	> 0.98
Asynchronous end-to-end latency (P95)	< 6 s
Auto-match rate (highest tier)	$> 60\%$ of scans
User correction rate	$< 15\%$

The final two measures, auto-match and correction rates, are the business-critical measures: they indicate whether the confidence tiers are calibrated correctly in production rather than merely on the test set.

Chapter 6

Discussion

6.1 Advantages of the Proposed Framework

The framework's principal advantage is **integration of the full chain**, spanning recognition, interpretation, resolution, regimen creation, and learning, in a system designed end-to-end for one market rather than assembled from generic parts. This integration yields several concrete benefits.

Accuracy under local conditions. Follows from the organizer's domain-specific steps (Bangla numeral normalization, the Bangladeshi abbreviation table, two-column prescription merging) that generic OCR systems lack.

Trustworthy automation. Follows from conservative confidence computation, taking the scan-level minimum across entries, tiered actions, and explicit escape paths, which ensures the system never silently attaches a wrong medicine.

Economic viability. Follows from the caching strategy and the staged hosting path (managed vision-language-model API initially, self-hosted open weights on GPU only when volume justifies it), keeping per-user marginal cost near zero for a product intended to be free.

Compound improvement. Follows from the active-learning loop, which makes the hardest part of the problem, the long tail of brand names, aliases, and handwriting styles, a solved-by-growth problem rather than a solved-by-labor problem.

Operational simplicity. Follows from the modular monolith with clean boundaries, which avoids premature microservice complexity while guaranteeing that extraction into independently scalable services is redeployment rather than re-engineering.

6.2 Limitations and Challenges

The framework carries identifiable limitations that deserve honest articulation.

Handwriting quality variance. Is the fundamental uncertainty: a 200-prescription evaluation set, however curated, cannot fully represent the diversity of practicing physicians' scripts, and severe cases may remain below acceptable accuracy.

Dataset staleness. Is a second-order risk: the base corpus, while comprehensive, reflects a snapshot of the market and must be refreshed on a scheduled cycle supplemented by regulatory and community sources.

Ambiguous abbreviations and dosages. Can produce clinically plausible but wrong interpretations (for example, confusing *OD* with *OS/OD* ophthalmic notation), which is why the confidence tiers exist but cannot fully eliminate the risk.

LLM cost and latency. Impose operational constraints: each scan consumes model inference, and latency budgets require careful mode dispatch between synchronous and asynchronous handling.

Offline-reconciliation edge cases. Medicine edits made on one device while another device's local schedule is mid-sync require the server-authoritative design to resolve, and residual complexity remains in caregiver multi-profile scenarios.

Regulatory positioning. Is deliberately consumer-grade: Nirog is a medication-management aid, not a diagnostic or prescribing device, and its outputs must never be presented as clinical advice.

6.3 Future Enhancements

Several extensions build naturally on the existing architecture. **Physician handwriting personalization** would fine-tune recognition on consenting users' repeat prescriptions, improving results for individual doctors over time. **Pharmacy and telemedicine integrations** would consume the resolved brand-generic graph for dispensing verification and structured documentation. **Patient education layers** would surface generic-class information, common indications, and drug interaction warnings alongside the medicine detail view. **Multilingual expansion** would generalize the pipeline to neighboring markets with similar brand fragmentation. **Model fine-tuning** on the growing corpus of annotated prescriptions and feedback would progressively reduce dependence on large general models. And a **web client** would extend caregiver access beyond the mobile application.

6.4 Clinical Safety and Edge Cases

Because the product operates in a health context, safety design is treated as a first-class requirement rather than an afterthought. The system adopts three defensive postures.

Never silent. Every uncertainty escalates to the user: the confidence tiers guarantee that low-confidence extractions are always surfaced, and the "None of these / Add new" path ensures the pipeline has a human-validated exit at every branch.

Conservative defaults. Scan-level confidence takes the minimum across entries, and the auto-attach threshold is set high enough that wrong attachments are rare events rather than common noise.

Human-in-the-loop curation. User-added medicines are invisible to other users until administrative verification, and the alias promotion loop requires consensus from three or more independent users before a correction enters the shared knowledge base.

Edge cases such as multi-page prescriptions, two-column layouts, mixed scripts, and retroactive dose logging are each handled by dedicated pipeline stages rather than by ad-hoc post-processing.

Chapter 7

Conclusion

7.1 Key Contributions

This whitepaper has presented Nirog, a framework that addresses the prescription-to-adherence gap in Bangladesh through a vertically integrated pipeline: OpenCV preprocessing of handwritten prescriptions, two-pass vision-language-model recognition faithful to the ink, domain-organized structured extraction with local medical abbreviation knowledge, a four-strategy semantic matching ensemble over a 21,000-medicine national corpus, conservative confidence tiering with explicit user-escape paths, and an active-learning feedback loop that compounds the knowledge base with every use. To our knowledge, no existing system connects handwritten prescription recognition in a mixed Bangla-English clinical register to a complete adherence automation loop with community-grown local drug knowledge, which constitutes the framework's principal contribution.

7.2 Impact and Applications

The immediate impact is patient-facing: lower-error medication records, on-time doses, refill awareness, and adherence measurement for patients and caregivers who today rely on memory and paper. The secondary impact is infrastructural: the verified brand-generic graph, the annotated prescription corpus, and the alias dictionary form reusable data assets for pharmacy systems, telemedicine documentation, insurance adjudication, and pharmacovigilance. The tertiary impact is methodological: the framework demonstrates that in markets where the prescription is handwritten and the drug landscape is locally branded, the correct architectural answer is not a better general-purpose OCR engine but a purpose-built chain of preprocessing, structured extraction, semantic resolution, and feedback-driven curation.

7.3 Future Directions

Future work extends in three directions. **Depth**: fine-tuning recognition on the accumulating annotated corpus and physician-specific handwriting to push character error rates further down. **Breadth**: extending the organizer and abbreviation schema to additional South Asian prescription conventions, and generalizing the matching ensemble to other nationally fragmented drug markets. **Integration**: connecting the resolved regimen graph to pharmacies, telemedicine platforms, and clinical decision support, so that a patient's digitally reconstructed prescription becomes the seed of a complete, longitudinal, machine-readable medication record.

Bibliography

[1] A. S. Sakib, "Assorted medicine dataset of bangladesh." https://www.kaggle.com/datasets/ahmedshahriarsakib/assorted-medicine-dataset-of-bangladesh, 2022. CC0 license; sourced from medex.com.bd; accessed August 2026.

[2] K. Patel *et al.*, "Mobile apps for increasing treatment adherence: Systematic review," *JMIR mHealth and uHealth*, vol. 7, no. 5, p. e12505, 2019.

[3] World Health Organization, "Adherence to long-term therapies: Evidence for action," tech. rep., World Health Organization, Geneva, 2003.

[4] Y. Peng, H. Wang, Q. Fang, L. Xie, L. Shu, W. Sun, and Q. Liu, "Effectiveness of mobile applications on medication adherence in adults with chronic diseases: A systematic review and meta-analysis," *Journal of Managed Care & Specialty Pharmacy*, vol. 26, no. 4, pp. 550–561, 2020.

[5] A. Mistry, D. Kellar, and H. Craig, "Do mobile device apps designed to support medication adherence yield efficacy? a systematic review of randomized controlled trials," *BMJ Open*, vol. 10, no. 1, p. e032045, 2020.

[6] B. Jiménez-Rodriguez *et al.*, "Medication adherence apps: Review and content analysis," *JMIR mHealth and uHealth*, vol. 6, no. 3, p. e62, 2018.

[7] V. Lanke, K. Trimm, B. Habib, and R. Tamblyn, "Evaluating the effectiveness of mobile apps on medication adherence for chronic conditions: Systematic review and meta-analysis," *Journal of Medical Internet Research*, vol. 27, p. e60822, 2025.

[8] F. Author and S. Author, "A novel approach of doctors' handwritten medical prescription recognition," *Journal of Data Governance*, 2026.

[9] F. Author and S. Author, "A deep learning-based framework for handwritten medical prescription recognition using ocr and nlp," in *ResearchGate Publication*, 2026.

[10] F. Author and S. Author, "Docpresrec: Doctor's handwritten prescription recognition using deep learning," in *Intelligent Document Processing*, CRC Press, 2022.

[11] F. Author and S. Author, "Intelligent prescription digitization: Leveraging multi-engine ocr and deep contextual nlp," in *IEEE Conference Proceedings*, 2025.

[12] J. S. 1807, "Medical-prescription-ocr (donut-based prescription ocr)." https://github.com/JonSnow1807/Medical-Prescription-OCR, 2024.

[13] F. Author and S. Author, "A hybrid approach to bangla handwritten ocr," *Discover Artificial Intelligence*, 2025.

[14] F. Author and S. Author, "Bilingual bangla ocr for rural empowerment," *IEEE Open Journal*, 2025.

[15] F. Author and S. Author, “A hypercomplex bangla handwriting character recognition dataset,” *PMC*, 2023.

[16] F. Author and S. Author, “Multi-head attention based interaction-aware architecture for bangla handwritten text recognition,” *arXiv preprint*, 2026.

[17] F. Author and S. Author, “Handwritten mixed-script recognition system: A comprehensive approach,” in *Proceedings of the 14th IAPR International Conference on Document Analysis and Recognition*, Springer, 2017.

[18] J. Chen *et al.*, “M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” *arXiv preprint*, 2024.

[19] BAAI, “Bge-m3: Multilingual embedding model.” https://huggingface.co/BAAI/bge-m3, 2024.

[20] B. E. Adventures, “Modular monoliths: The architecture pattern we don’t talk enough about,” 2023.

[21] Z. Wu *et al.*, “Deepseek-vl2: Mixture-of-experts vision-language models for advanced multimodal understanding,” *arXiv preprint*, 2024.

[22] F. Team, “Guide to app architecture.” https://docs.flutter.dev/app-architecture/guide, 2024.

[23] Google, “Firebase cloud messaging.” https://firebase.google.com/products/cloud-messaging, 2024.

[24] M. Guns, “flutter_local_notifications.” https://pub.dev/packages/flutter_local_notifications, 2024.

[25] M. Li *et al.*, “Trocr: Transformer-based optical character recognition with pre-trained models,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.